REVISION TEST -I

CLASS : 12TH STD. **MARKS : 70**

SUBJECT : CHEMISTRY **TIME : 2.30HRS**

PART-I

Chosse the correct answers. **15 X 1 =15**

1. Zinc is obtained from ZnO by
   a) Carbon reduction b) Reduction using silver
   c) Electrochemical process d) Acid leaching
2. Thermodynamically the most stable form of carbon is
   a) Diamond b) graphite c) Fullerene d) none of these
3. Which is true regarding nitrogen?
   a) least electronegative element b) has low ionisation enthalpy than oxygen
   c) d- orbitals available d) ability to form p p π π − bonds with itself
4. Which transition element shows the highest oxidation State
   a)Os b)Ti c)Sc d)Zn
5. IUPAC name of the complex $K_3[Al(C_2O_4)_3]$
   a) potassiumtrioxalatoaluminium(III) b) potassiumtrioxalatoaluminate(II)
   c) potassiumtrisoxalatoaluminate(III) d) potassiumtrioxalatoaluminate(III)
6. Solid NH_3 is an example of
   a)Non polar molecular solid b)polar molecular solid
   c) hydrogen bonded molecular solid d) metallic solid
7. For a reaction the rate law is rate=$K[A]^{6/2}[B]^{-4/2}$the overall order of reaction is
   a)1 b)2 c)6/2 d)-4/2
8. . Which of these is not likely to act as lewis base?
   a) BF_3 b) PF_3 c) CO d) F^-
9. Faradays constant is defined as
   a) charge carried by 1 electron
   b) charge carried by one mole of electrons
   c) charge required to deposit one mole of substance
   d) charge carried by 6.22 X 10^{-10} electrons
10. Which one of the following is an example for homogeneous catalysis?
    a) manufacture of ammonia by Haber's process
    b) manufacture of sulphuric acid by contact process
    c) hydrogenation of oil d) Hydrolysis of sucrose in presence of all HCl
11. Oxidation of glycerol with con HNO_3 gives to
    a) Glyceric acid b)oxalic acid c)mesooxalic acid d)tatronic acid

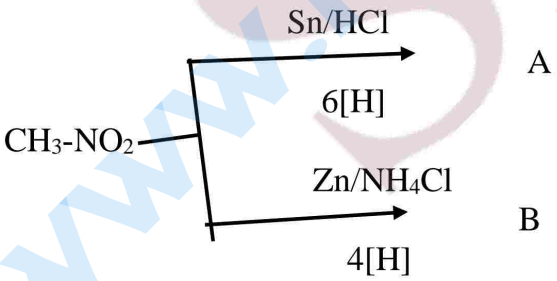
12. On reacting with neutral ferric chloride, phenol gives
    a) red colour b) violet colour c) dark green colour d) no colouration.

13. Which one of the following reaction is an example of disproporationation reaction
    a) Aldol condensation b) cannizaro reaction
    c) Benzoin condensation d) none of these
14. Which of the following reagent can be used to convert nitrobenzene to aniline
    a) Sn / HCl b) ZnHg / NaOH c) $LiAlH_4$ d) All of these
15. . Which one given below is a non-reducing sugar?
    a) Glucose b) Sucrose c) maltose d) Lactose

PART-II

Answer the following any six questions. 6 X 2 = 12
Question number 24 is compulsory.

16. What are interhalogen compounds ? give two examples
17. Out of $Lu(OH)_3$ and $La(OH)_3$ which is more basic and why ?
18. Write the IUPAC names for the following compounds
    i) $[Cu(NH_3)_4]SO_4$ ii) $Na_2[Ni(EDTA)]$
19. Define packing efficiency
20. What are conjugate acid base pair give an example ?
21. Write a note on tyndall effect?
22. what is metamerism and give one example
23. Write the rosenmud reduction?
24. From the following reaction identify A and B .



PART-III

Answer the following any six questions. 6 X 3 = 18
Question number 33 is compulsory.

25. what are the factors responsible for the anomalous behaviour of first element of the p-block ?
26. What is lanthanoide contraction explain its consequences
27. write a short note on postulates of werner's theory?

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28. Derive the Relation between P^H and P^{OH}

29. State faraday's law of electrolysis ?

30 Write any three characters of catalysts ?

31.Differentiate primary secondary and tertiary alcohols using Lucas test

32. .Write a short note on peptide bond ?

33. Write the mechanism of aldol condensation reaction ?

PART-IV

Answer the following all questions. **5 X 5 = 25**

34.a) i) Describe mond process for refining nickel.(3)

ii) How will you identify borate radical ? (2)

(OR)

b) i) Write the balanced equation for the overall reaction of chlorine with cold. NaOH and hot NaOH (2)

ii) What are interstitial compounds? Give example (3)

35.a) $[Ni(CN)_4]^{2-}$ is diamagnetic , while $[NiCl_4]^{2-}$ is paramagnetic explain (5)

(OR)

b) i) Classify molecular crystal with an example for each type.(3)

ii) Explain hume -rothery rule for formation of alloys ? (2)

36.a) i) Derive an expression for ostwald dilution law(3)

ii) Ionic conductance at infinite dilution of Al^{3+} and SO_4^{2-} 189 and 160 mho cm^2 equivl calculate the equivalent and molar conductance of the electrolyte $Al_2(SO_4)_3$ at infinite dilution. (2)

(OR)

b) i) Derive integrated rate law for a first order reaction A →product (3)

ii)What is Baeyer's regent? Write down its use ?(2)

37.a) i) How are the following conversion effected?

1) ethylene glycol → 1,4 dioxane(1 ½)

2) glycerol → acrolein(1 ½)

ii) How does ammonia react with formaldehyde ? Draw the structure of products (2)

(OR)

b) I)Explain the mechanism of cannizaro reaction?(3)

ii) Give the any two test for carboxylic acid group(2)

38. a) i) What is isoelectric point?(2)

ii) Mention the importance of carbohydrate(3)

(OR)

b) i) write short note on Gabriel phthalimide synthesis (3)

ii) Write the zwitter ion structure of alanine? (2)